UTM UNIVERSITI TEKNOLOGI MALAYSIA — Sekolah Pendidikan Profesional dan Pendidikan Berterusan (UTMSPACE)	Tutorial 1
	CHEMISTRY II
	FSPC 0034

1. A bombardier beetle (*Brachinus*) adopt a less passive defense mechanism. When threatened, the beetle squeezes some fluid from the inner compartment into the outer compartment, where in the presence of the enzymes, an exothermic reaction takes place:

$$C_6H_4(OH)_2\ (aq)\ +\ H_2O_2\ (aq) \rightarrow C_6H_4O_2\ (aq)\ +\ 2\ H_2O\ (l)$$

   Calculate the enthalpy of reaction from the given equations below:

   $C_6H_4(OH)_2\ (aq) \rightarrow C_6H_4O_2\ (aq) + H_2(g)$ $\Delta H^o = 177$ kJ/mol
   $H_2O_2\ (aq) \rightarrow H_2O\ (l) + ½\ O_2\ (g)$ $\Delta H^o = -94.6$ kJ/mol
   $H_2\ (g) + ½\ O_2\ (g) \rightarrow H_2O\ (l)$ $\Delta H^o = -286$ kJ/mol

2. A 1.50 dm^3 sample of a mixture of ethane gas, C_2H_6 and oxygen gas, measured at 25 ^oC and 101 kPa, was allowed to react in a bomb calorimeter in which, had a heat capacity of 5.03 kJ/K altogether with its contents. The complete combustion of the ethane gas to carbon dioxide gas and water caused a temperature rise in the calorimeter of 6.18 K. Given that ΔH^o_c (C_2H_6) is -1560 kJ/mol.

   a. Define the standard enthalpy of combustion of ethane.
   b. Write a thermochemical equation for the combustion of ethane.
   c. Knowing the ΔH^o_c of ethane, calculate the number of mole of ethane in the mixture.
   d. Calculate the total moles of gases in the calorimeter.

3. Cyclohexene, C_6H_{10} undergoes hydrogenation to produce cyclohexane, C_6H_{12} according to the equation given below:

$$C_6H_{10}\ (l)\ +\ H_2\ (g) \rightarrow C_6H_{12}\ (l)$$
$$\Delta H_f^o(C_6H_{10}) = -30 \text{ kJ/mol}$$
$$\Delta H_f^o(C_6H_{12}) = -3930 \text{ kJ/mol}$$

   a. Calculate the standard enthalpy of hydrogenation, ΔH^o of cyclohexene.
   b. Sketch the enthalpy diagram of this reaction

4. A quantity of 1.435 g of naphthalene, $C_{10}H_8$, a pungent-smelling substance used in moth repellents, was burned in a constant-volume bomb calorimeter. The temperature rose from 20.28°C to 25.95°C. If the heat capacity of the bomb plus water was 10.17 kJ/^{o}C, calculate the molar heat of combustion of naphthalene.

5. A copper pellet having a mass of 26.47 g at 89.98°C was placed in a constant-pressure calorimeter containing 100.0 mL of water. The water temperature increase from 22.50^{o}C

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to 23.17°C. Assuming that no heat is lost to the surroundings, calculate the specific heat of the copper pellet. [s(H_2O) = 4.184 J/g.°C and d(H_2O) = 1 g/mL]

6. Butane is a hydrocarbon gas that commonly used for cooking. The standard enthalpy of combustion of butane is -220.1 kJ/mol. 10 g of butane is used to heat 8 L of water. Assuming that no heat is lost to the surroundings, what is the increase in the water's temperature? Given that the specific heat capacity of water is, s(H_2O) = 4.18 J/g.°C and water density is 1 g/mL.

7. Given the following data, determine the enthalpy change for the formation of 40 g of methane, CH_4.

$$C\,(s) + 2\,H_2\,(g) \rightarrow CH_4\,(g)$$

$$CH_4\,(g) + 2\,O_2\,(g) \rightarrow CO_2\,(g) + 2\,H_2O\,(l) \qquad \Delta H_c^o = -890.3\text{ kJ/mol}$$

$$C\,(s) + O_2\,(g) \rightarrow CO_2\,(g) \qquad \Delta H_f^o = -393.5\text{ kJ/mol}$$

$$H_2\,(g) + \tfrac{1}{2}\,O_2\,(g) \rightarrow H_2O\,(l) \qquad \Delta H_f^o = -285.8\text{ kJ/mol}$$

8. Determine whether each process is exothermic or endothermic:
   a. Hand sanitizer evaporating from skin
   b. Hot pack used as muscle pain reliever

9. Calculate the enthalpy change of the reaction;

$$CH_4\,(g) + Br_2\,(g) \rightarrow CH_3Br\,(g) + HBr\,(g),$$

using the bond energies give in Table 1.

Table 1. Average bond enthalpies

bond	**ΔH_b/kJmol^{-1}**
C-H	+413
C-Br	+280
Br-Br	+193
H-Br	+366

10. Magnesium chloride, $MgCl_2$, is an ionic compound used to treat magnesium deficiency in the body.
    a. Complete this Born−Haber cycle for magnesium chloride by giving the missing species on the dotted lines. Include state symbols where appropriate. The energy levels are not drawn to scale.

1) $C_6H_4(OH_2)(aq) \rightarrow C_6H_4O_2(aq) + \cancel{H_2(g)}$ $\Delta H^\circ = 177$

$H_2O_2(aq) \rightarrow H_2O(l) + \cancel{\tfrac{1}{2}O_2(g)}$ $\Delta H^\circ = -94.6$

$\cancel{H_2(g)} + \cancel{\tfrac{1}{2}O_2(g)} \rightarrow H_2O(l)$ $\Delta H^\circ = -286$

$C_6H_4(OH_2) + H_2O_2 \rightarrow C_6H_4O_2 + 2H_2O$ $\Delta H^\circ = -203.6$

2) $V = 1.5\,dm^3$, $T = 25°C$, $P = 101000\,Pa$, $\Delta H^\circ_c\, C_2H_6 = -1560\,kJ$, $C = 5.03\,kJ/K$

(a) the enthalpy change when 1 mol of ethane completly burn in excess oxygen to produce carbon dioxide and water

(b) $C_2H_{6}g + \frac{7}{2}O_{2}g \rightarrow 2CO_{2}g + 3H_2O(l)$

(c) $\Delta H^\circ_c = \frac{-q}{mol}$ $q = C_p \times \Delta T$ (bomb)

$1560 = \frac{q}{mol}$ $= 5.030 \times 6.18$

$1560 = \frac{31.09\,kJ}{mol}$ $= 31.09\,kJ$

$mol = 0.019\,mol$

(d) 1 mol of C_2H_6 gas will react with 3.5 mol of O_2

0.019 mol " 0.07 mol "

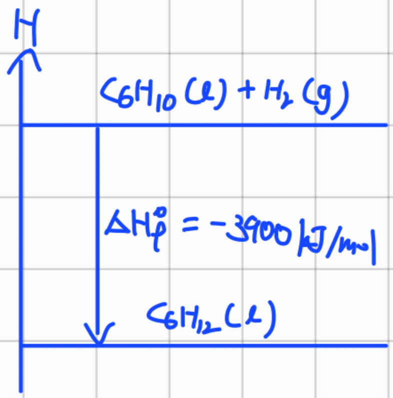
Total mol = 0.019 + 0.07 = 0.089 mol

3) (a) $\Delta H^\circ = \Delta H^\circ product - \Delta H^\circ reactant$

$= -3930 - (-30 + 0)$

$= -3900\,kJ/mol$

(b)



4) $q = C_p \times \Delta T$

$= 10.17\,(25.95 - 20.28)$

$= 57.66\ kJ$

molar heat of combustion : heat released when one mole of a substance is combusted

$mol = \frac{mass}{m.m.} = \frac{1.435}{10(12) + 8(1)} = 0.0112\ mol$

$\Delta H° = \frac{-q}{mol} = \frac{-57.66}{0.0112} = -5148.21\ kJ/mol$

5) $ms\Delta T = ms\Delta T$

$26.47\,(s)(89.98 - 23.17) = \left(100\,ml \times \frac{1g}{ml}\right)(4.184)(23.17 - 22.50)$

$1768.46\,s = 280.33$

$s = 0.159\ J/g.°C$

6) $\Delta H° = -220.1\ kJ/mol$, $m = 10g$, $V = 8\ell$

mass of water $= 8\cancel{\ell} \times \frac{1000\,\cancel{ml}}{1\,\cancel{\ell}} \times \frac{1g}{1\,\cancel{ml}} = 8000g$

$-220.1 = \frac{-q}{mol}$

$220.1 = \frac{q}{mol}$

$q = 220.1 \times \frac{10}{58}$

$q = 37.95\ kJ$

$ms\Delta T = ms\Delta T$

$37950 = 8000(4.18)(\Delta T)$

$\Delta T = 1.135°C$

7) $\Delta H° = ?$, $m = 40g$, $m.m. = 16$

$mol = \frac{40}{16} = 2.5\ mol$

$\cancel{CO_2}(g) + \cancel{2H_2O}(\ell) \rightarrow CH_4 + \cancel{2O_2}$ $\quad \Delta H_c° = 890.3\ kJ/mol$

$C(s) + \cancel{O_2}(g) \rightarrow \cancel{CO_2}(g)$ $\quad \Delta H_f° = -393.5\ kJ/mol$

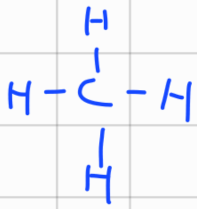
$2H_2 + \cancel{O_2}(g) \rightarrow \cancel{2H_2O}(\ell)$ $\quad \Delta H_f° = 2(-285.8)\ kJ/mol$

$\Delta H_f° = -74.8\ kJ/mol$

8) (a) endothermic

(b) exothermic

9)

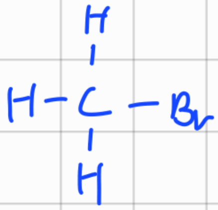


Br-Br

$4\Delta H_{bond}(C-H) = 4(+413) = 1652$

$1\Delta H_{bond}(Br-Br) = +193$

$\Sigma H_{bond} = +1845\ kJ/mol$



H-Br

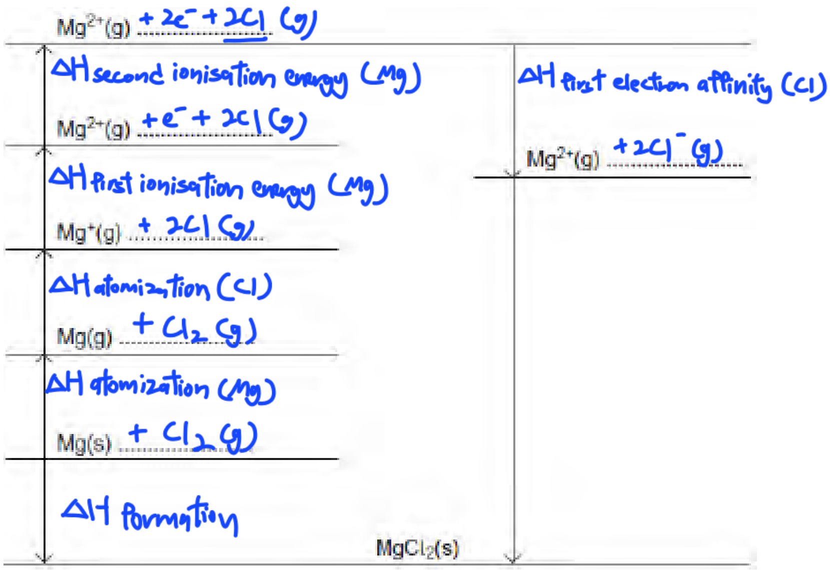
$3\Delta H_{bond}(C-H) = 3(+413)$

$1\Delta H_{bond}(C-Br) = +280$

$1\Delta H_{bond}(H-Br) = +366$

$\Sigma H_{bond} = +1885\ kJ/mol$

$$\Delta H_f = \Delta H_a - \Delta H_{bond}$$
$$= +1885 - (+1845)$$
$$= +40\ kJ/mol$$



b. Use your Born–Haber cycle from part (a) and data from Table 2 to calculate the value for the electron affinity of chlorine.

Table 2. Enthalpy change in a Born-Haber cycle.

Enthalpy Change	Enthalpies (ΔH/kJmol^{-1})
Enthalpy of atomisation of magnesium	+150
Enthalpy of atomisation of chlorine	+121
First ionisation energy of magnesium	+736
Second ionisation energy of magnesium	+1450
Enthalpy of formation of magnesium chloride	-642
Lattice enthalpy of formation of magnesium chloride	-2493

10)(b)

$Mg(s) \rightarrow Mg(g)$ $\quad \Delta H = +150$

$Cl_2(s) \rightarrow 2Cl(g)$ $\quad \Delta H = 2(+121) = +242$

$Mg(g) \rightarrow Mg^{+}(g) + e^-$ $\quad \Delta H = +736$

$Mg^{+}(g) \rightarrow Mg^{2+}(g) + e^-$ $\quad \Delta H = +1450$

$Mg^{2+}(g) + Cl^-(g) \rightarrow MgCl_2(s)$ $\quad \Delta H = -2493$

$2Cl^-(g) \rightarrow 2Cl(g) + 2e^-$ $\quad \Delta H = 2x$

$Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$ $\quad \Delta H = -642$

$150 + 242 + 736 + 1450 - 2493 + 2x = -642$

$2x = -727$

$x = -363.5$ kJ/mol